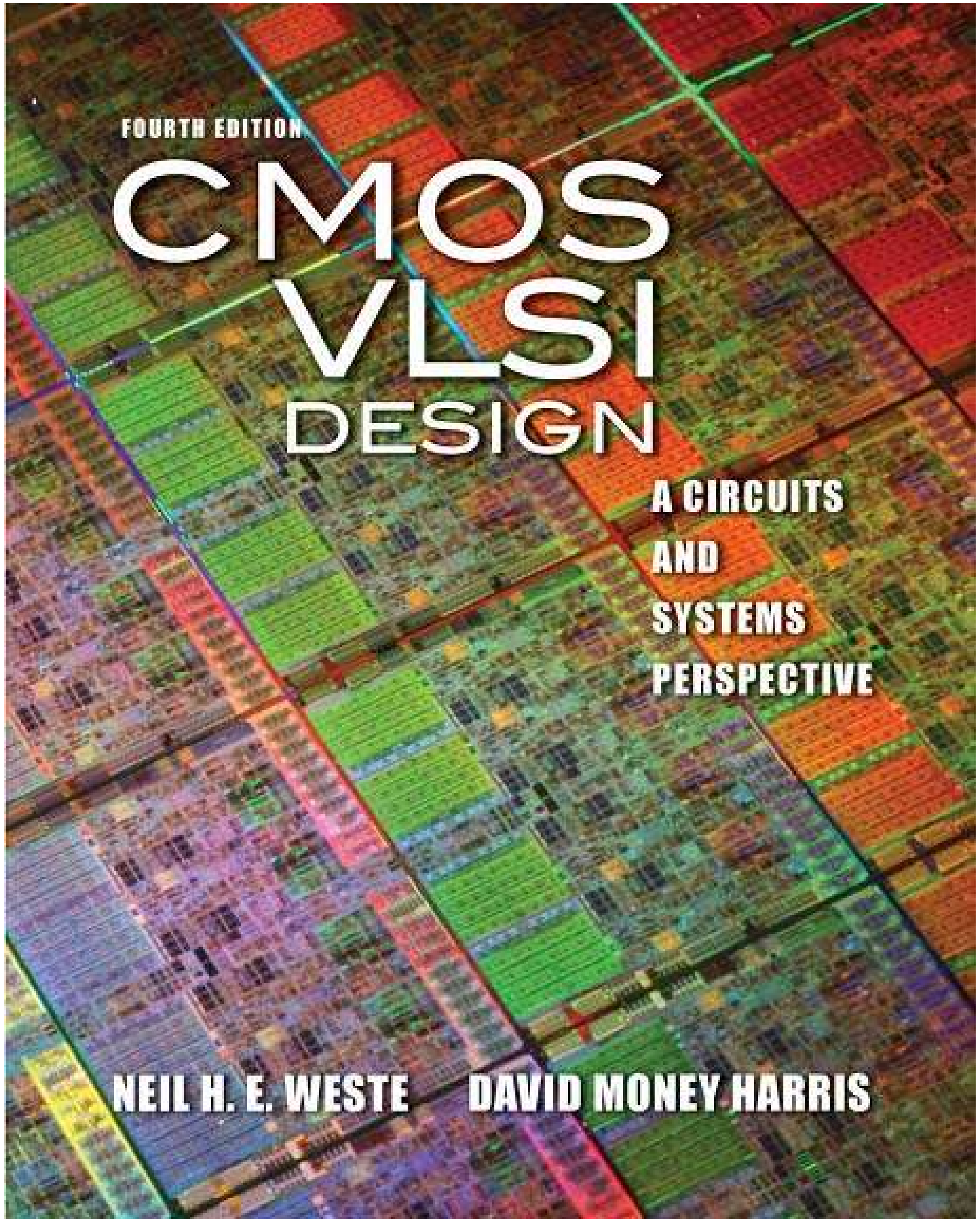


Lecture 6: Circuit design – part 2

6.1 Combinational circuit design

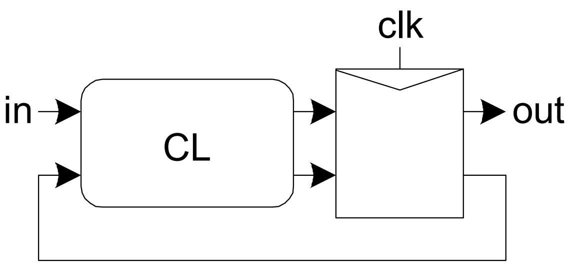
6.2 Sequential circuit design

6.3 Circuit simulation

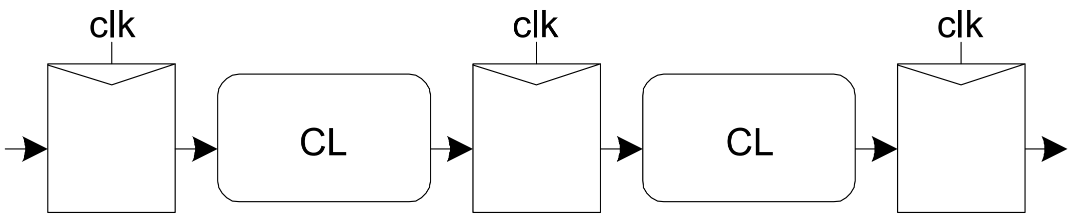
6.4. Hardware description language

2. Sequential circuit design

- *Combinational logic*
  - output depends on **current inputs**
- *Sequential logic*
  - output depends on **current and previous inputs**
  - Requires separating previous, current, future
  - Called *state* or *tokens*
  - Ex: FSM, pipeline



Finite State Machine



Pipeline

Sequencing Cont.

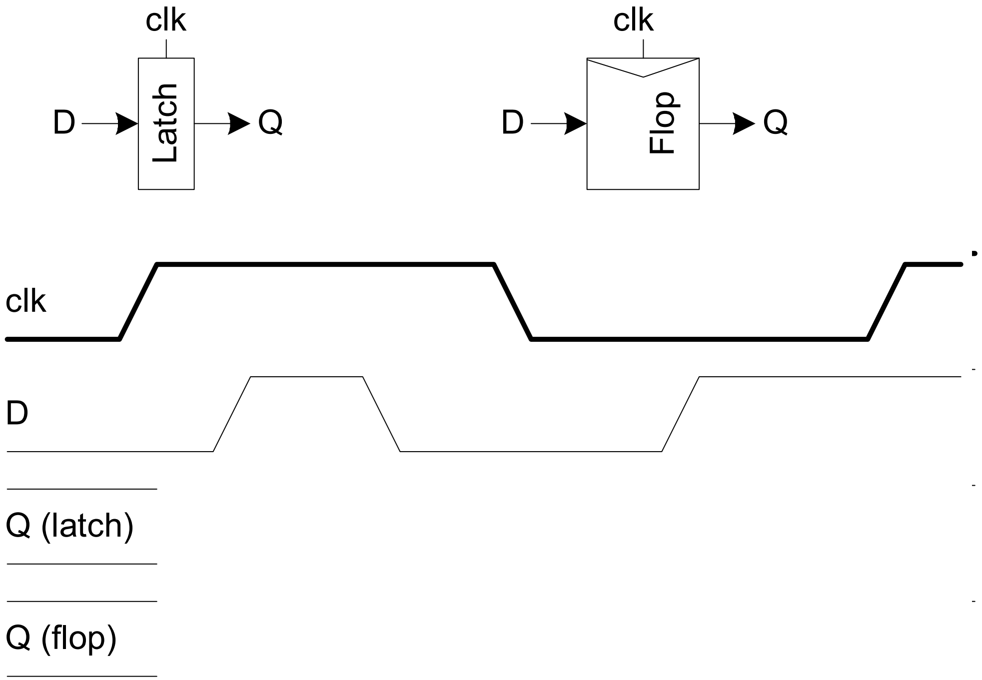
- If tokens moved through pipeline at constant speed, no sequencing elements would be necessary
- Ex: fiber-optic cable
  - Light pulses (tokens) are sent down cable
  - Next pulse sent before first reaches end of cable
  - No need for hardware to separate pulses
  - But *dispersion* sets min time between pulses
- This is called *wave pipelining* in circuits
- In most circuits, dispersion is high
  - Delay fast tokens so they don't catch slow ones.

Sequencing Overhead

- Use flip-flops to delay fast tokens so they move through exactly one stage each cycle.
- Inevitably adds some delay to the slow tokens
- Makes circuit slower than just the logic delay
  - Called sequencing overhead
- Some people call this clocking overhead
  - But it applies to asynchronous circuits too
  - Inevitable side effect of maintaining sequence

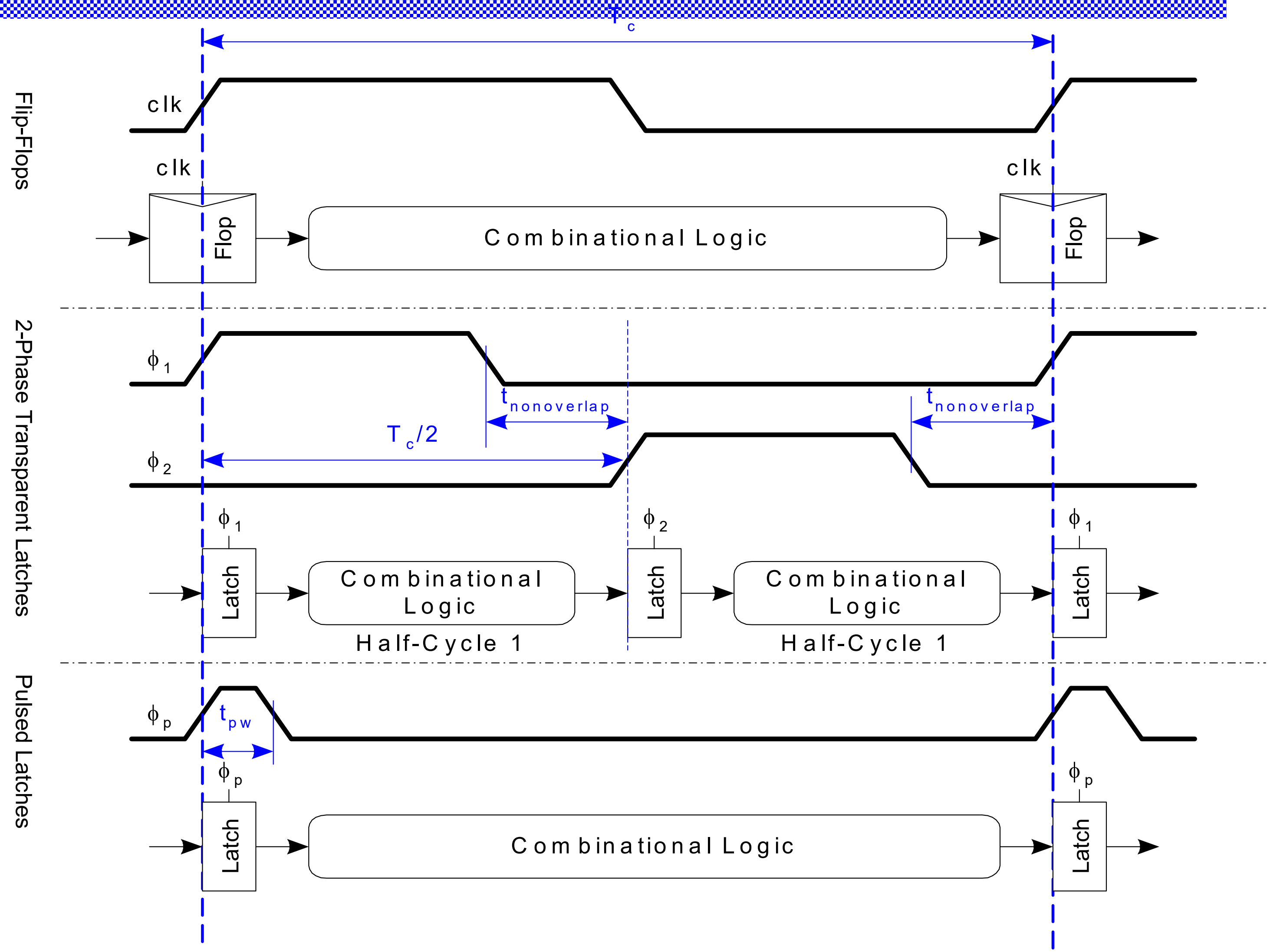
Sequencing Elements

- **Latch**: Level sensitive
  - a.k.a. (also known as) transparent latch, D latch
- **Flip-flop**: edge triggered
  - A.k.a. master-slave flip-flop, D flip-flop, D register
- Timing Diagrams
  - Transparent
  - Opaque
  - Edge-trigger



Sequencing Methods

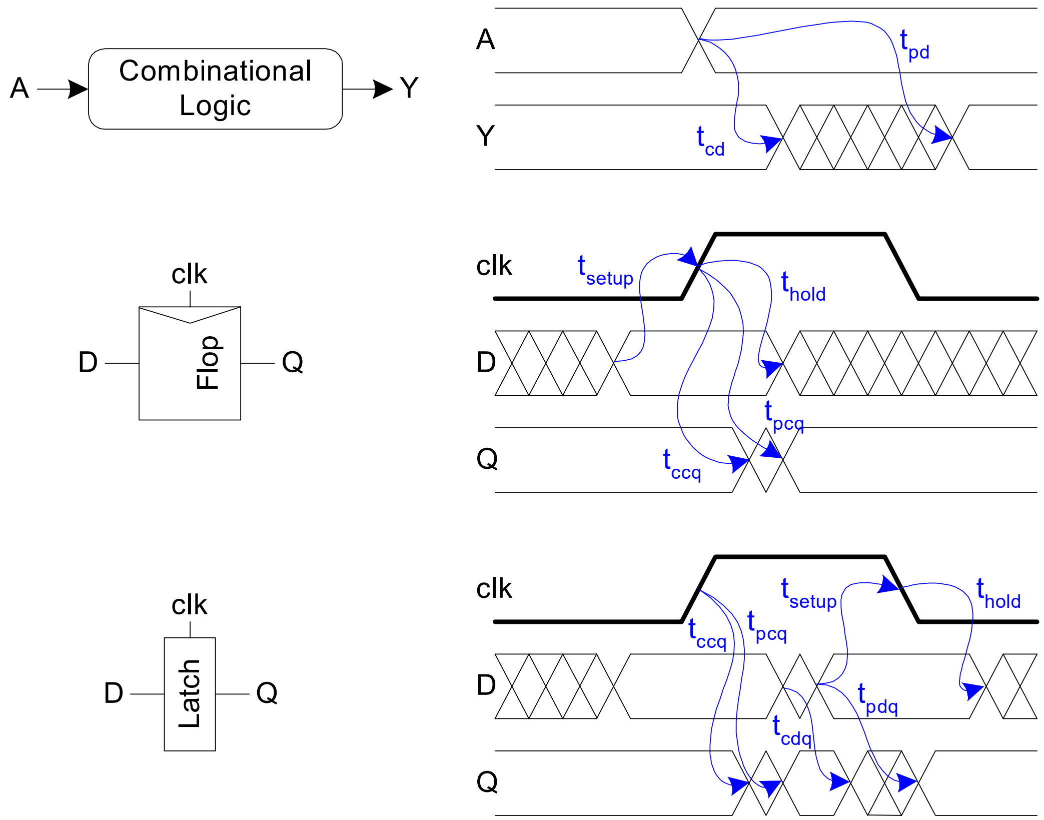
- Flip-flops
- 2-Phase Latches
- Pulsed Latches



Timing Diagrams

Contamination and Propagation Delays

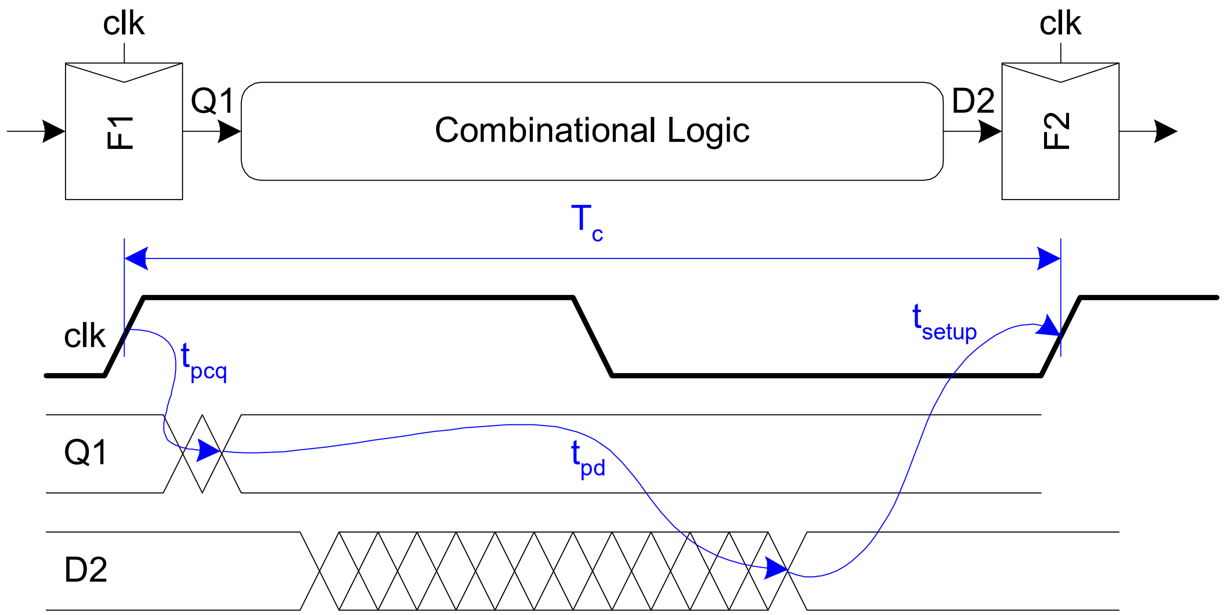
t_{pd}	Logic Prop. Delay
t_{cd}	Logic Cont. Delay
t_{pcq}	Latch/Flop Clk->Q Prop. Delay
t_{ccq}	Latch/Flop Clk->Q Cont. Delay
t_{pdq}	Latch D->Q Prop. Delay
t_{cdq}	Latch D->Q Cont. Delay
t_{setup}	Latch/Flop Setup Time
t_{hold}	Latch/Flop Hold Time



Max-Delay Constraints: Flip-Flops

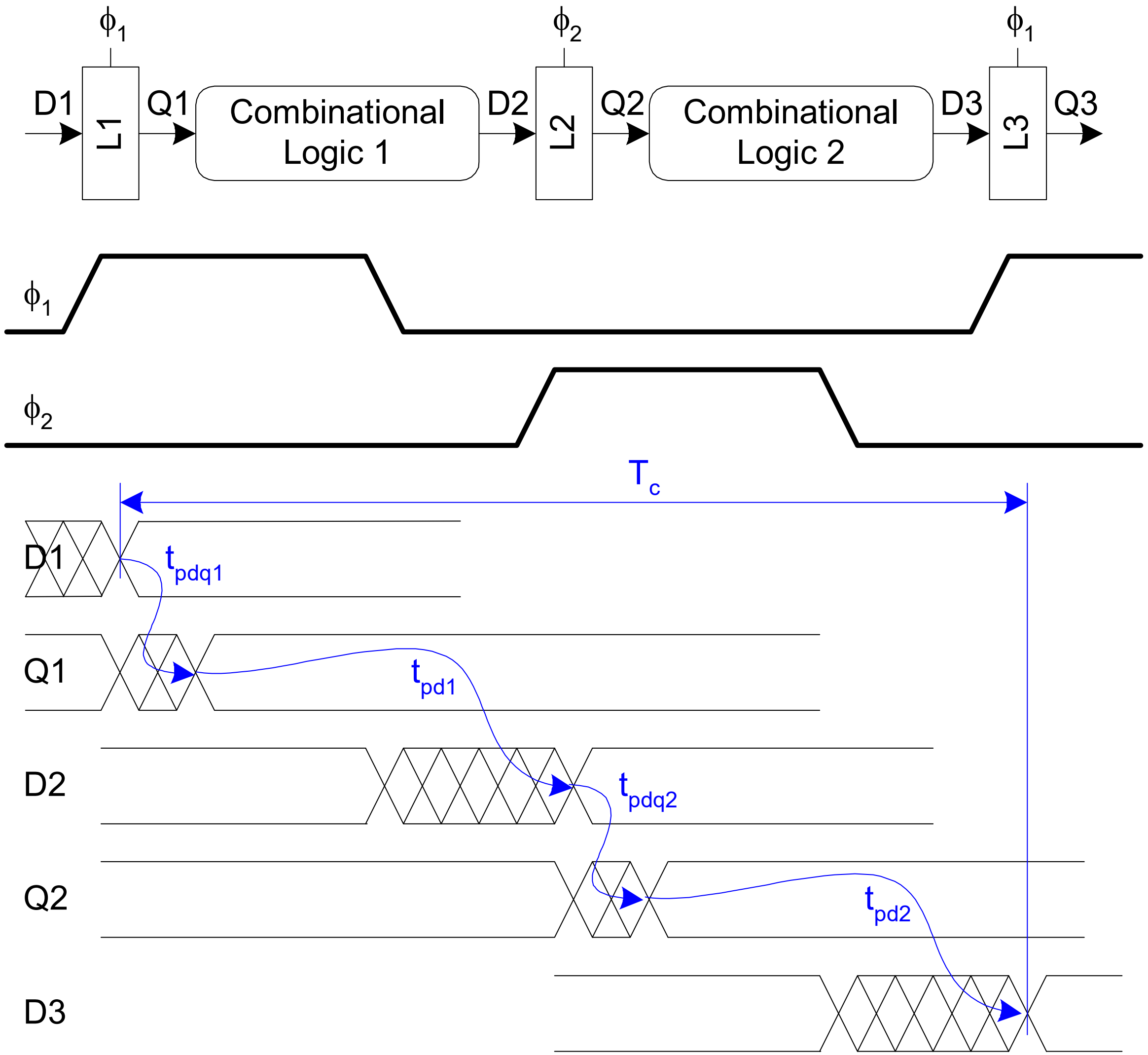
- If the combination logic delay is too great, the receiving element will miss its setup time and sample the wrong value.
- This is called a ***setup time failure*** or ***max-delay failure.***
- It can be solved by redesigning the logic to be faster or by increasing the clock period.

$$t_{pd} \le T_c - \underbrace{\left(\qquad\qquad \right)}_{\text{sequencing overhead}}$$



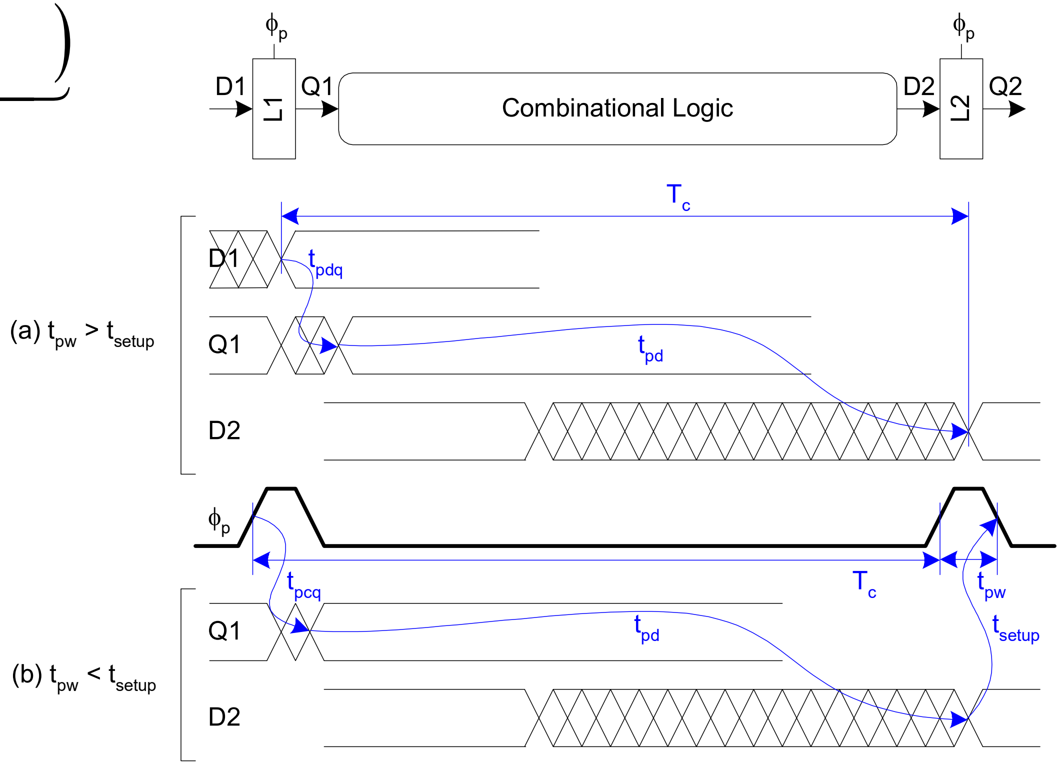
Max Delay Constraints : 2-Phase Latches

$$t_{pd} = t_{pd1} + t_{pd2} \leq T_c - \underbrace{(\quad\quad)}_{\text{sequencing overhead}}$$



Max Delay: Pulsed Latches

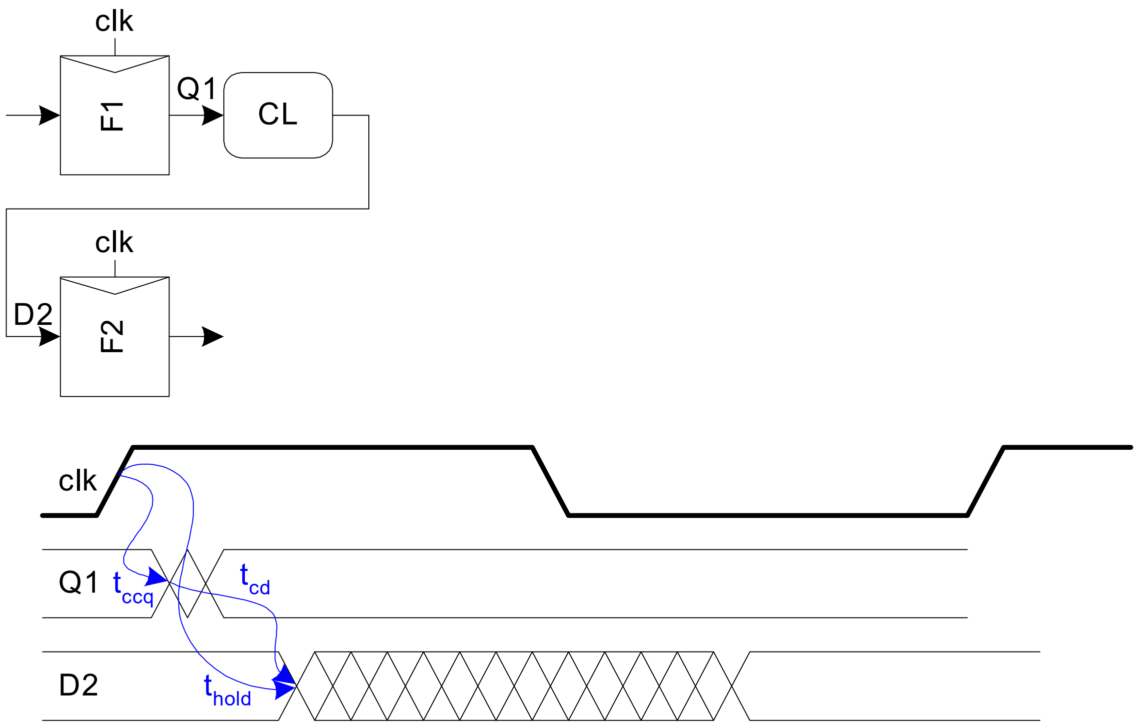
$$t_{pd} \le T_c - \underbrace{\max\left(\qquad\qquad\qquad \right)}_{\text{sequencing overhead}}$$



Min-Delay Constraints: Flip-Flops

- If the hold time is large and the contamination delay is small, data can incorrectly propagate through two successive elements on one clock edge, corrupting the state of the system.
- This is called a *race condition*, *hold-time failure*, or *min-delay failure*.
- It can only be fixed by redesigning the logic, not by slowing the clock.

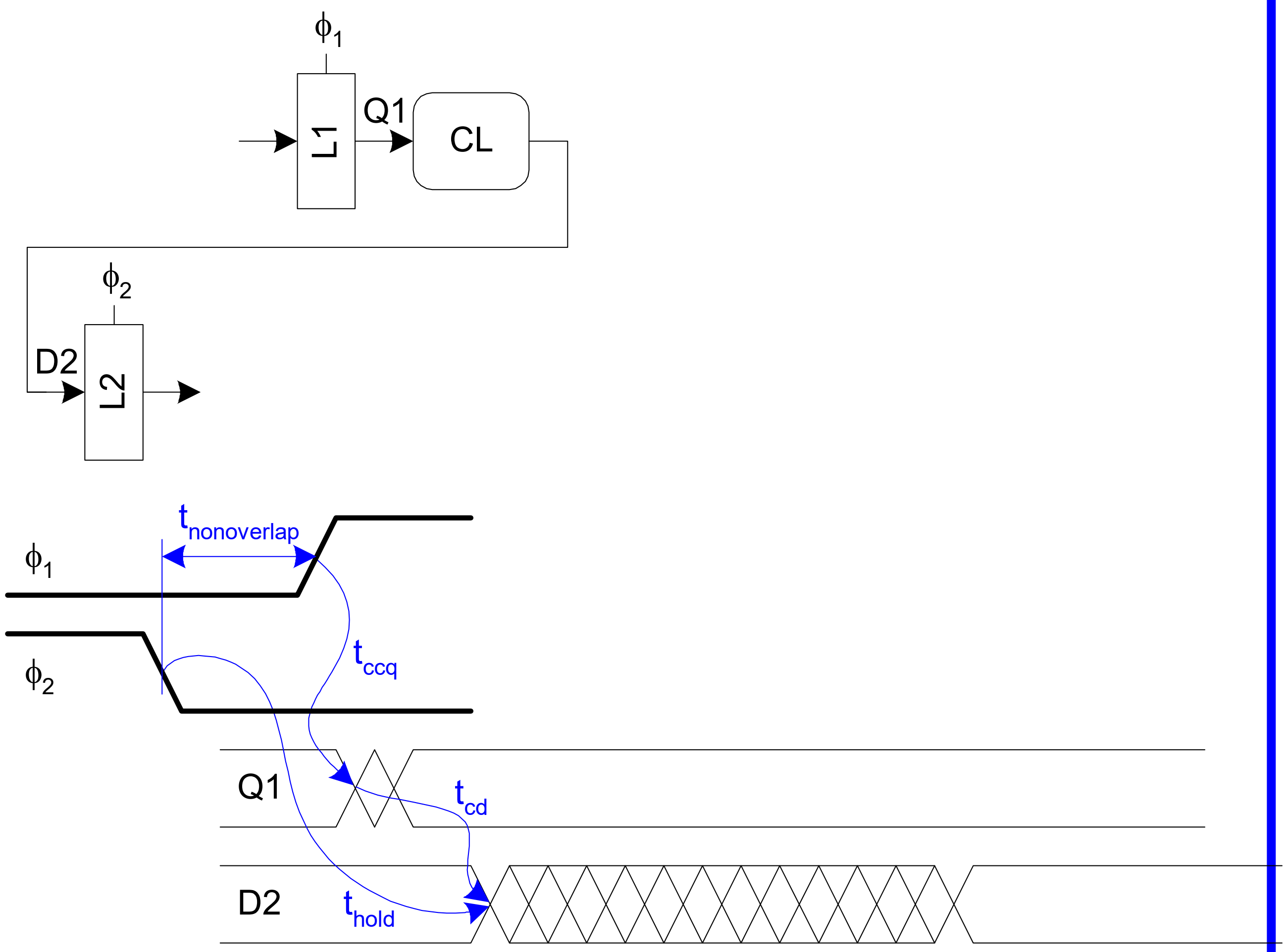
$t_{cd} \geq$



Min-Delay Constraints : 2-Phase Latches

$$t_{cd1}, t_{cd2} \geq$$

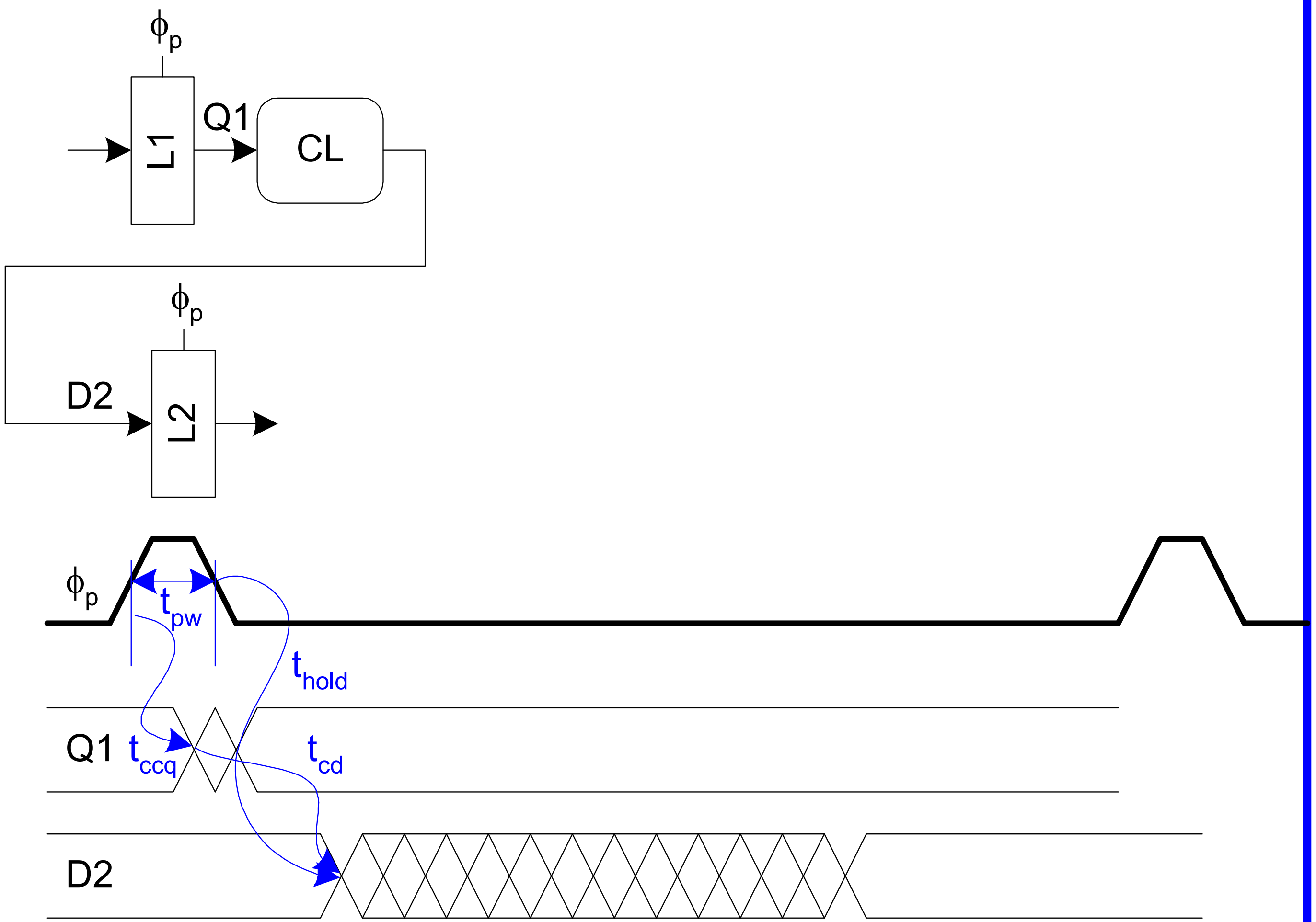
- Hold time reduced by nonoverlap
- Paradox: hold applies twice each cycle, vs. only once for flops.
- But a flop is made of two latches!



Min-Delay Constraints: Pulsed Latches

$t_{cd} \geq$

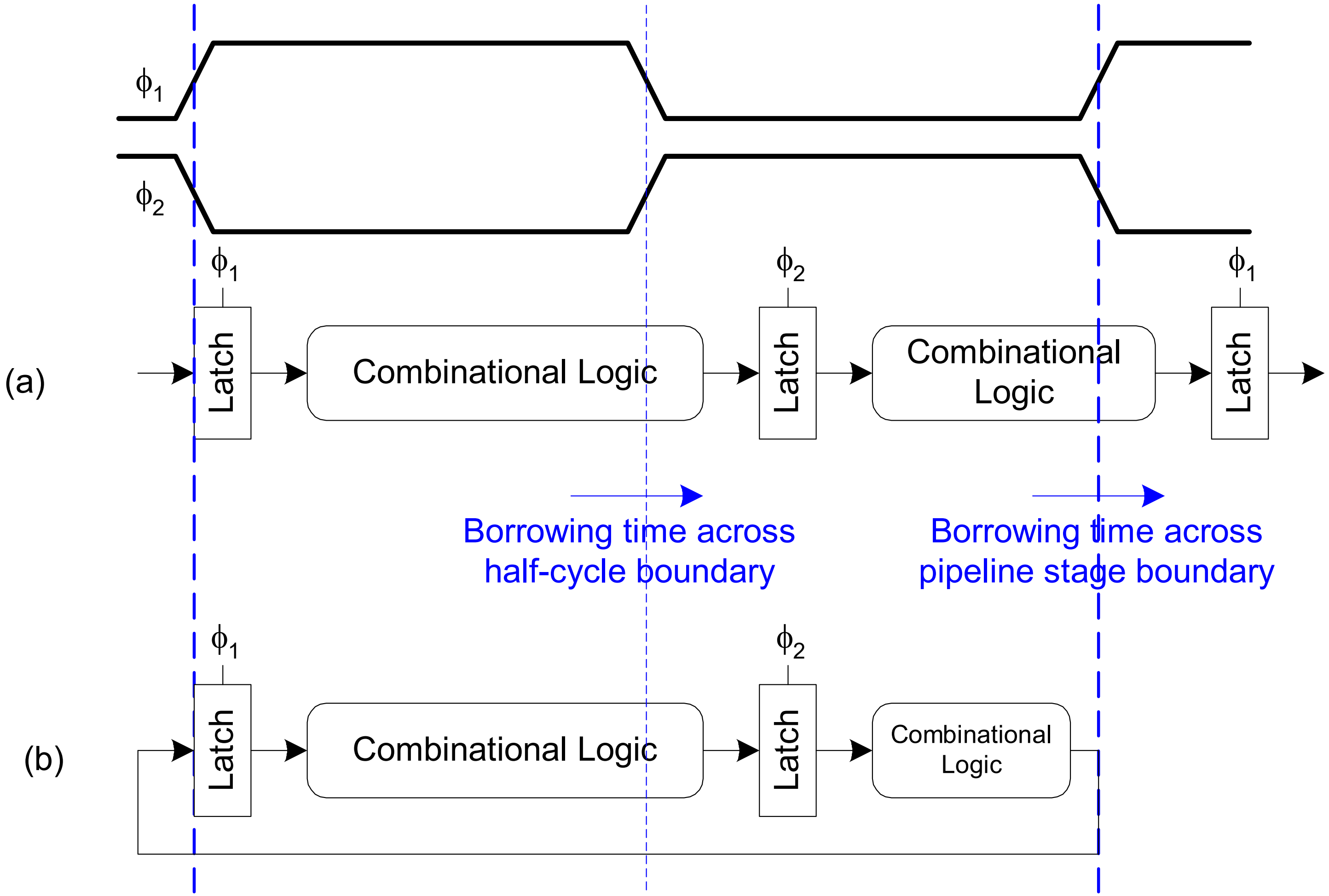
Hold time increased by pulse width



Time Borrowing

- In a flop-based system:
  - Data launches on one rising edge
  - Must setup before next rising edge
  - If it arrives late, system fails
  - If it arrives early, time is wasted
  - Flops have hard edges
- In a latch-based system
  - Data can pass through latch while transparent
  - Long cycle of logic can borrow time into next
  - As long as each loop completes in one cycle

Time Borrowing Example



Loops may borrow time internally but must complete within the cycle

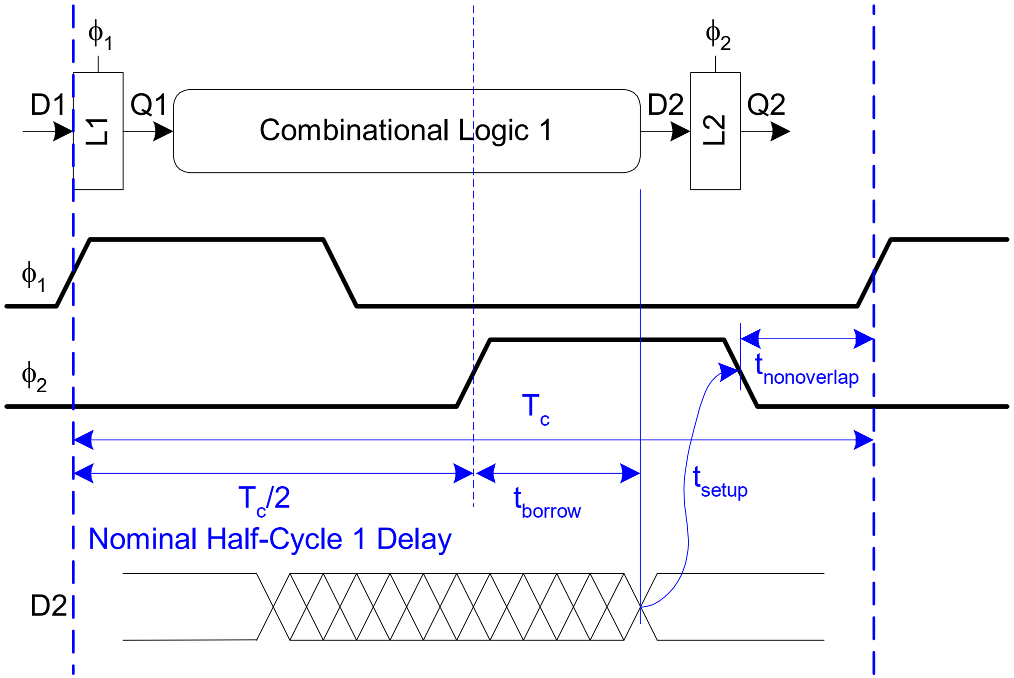
How Much Borrowing?

2-Phase Latches

$$t_{\text{borrow}} \leq \frac{T_c}{2} - \left(t_{\text{setup}} + t_{\text{nonoverlap}}\right)$$

Pulsed Latches

$$t_{\text{borrow}} \leq t_{pw} - t_{\text{setup}}$$



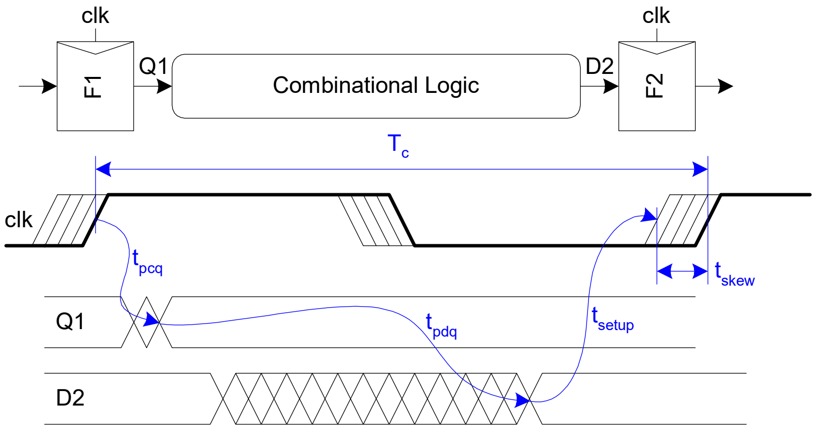
Clock Skew

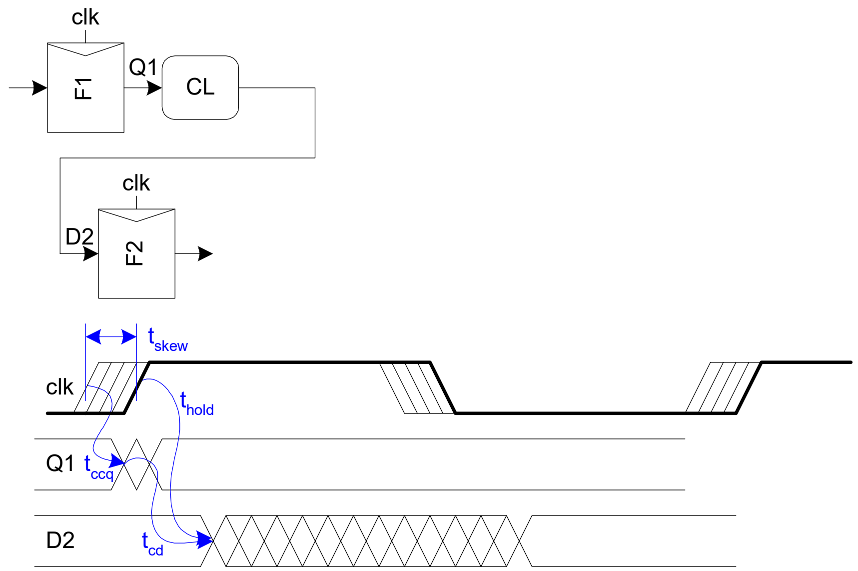
- **Clock skew** is a phenomenon in which the same sourced clock signal arrives at different components (generally latches or flip-flops) at different times
- We have assumed ideal clocks with zero skew
- Clocks really have uncertainty in arrival time
  - Decreases maximum propagation delay
  - Increases minimum contamination delay
  - Decreases time borrowing

Skew: Flip-Flops

$$t_{pd} \le T_c - \underbrace{\left(t_{pcq} + t_{\text{setup}} + t_{\text{skew}}\right)}_{\text{sequencing overhead}}$$

$$t_{cd} \ge t_{\text{hold}} - t_{ccq} + t_{\text{skew}}$$





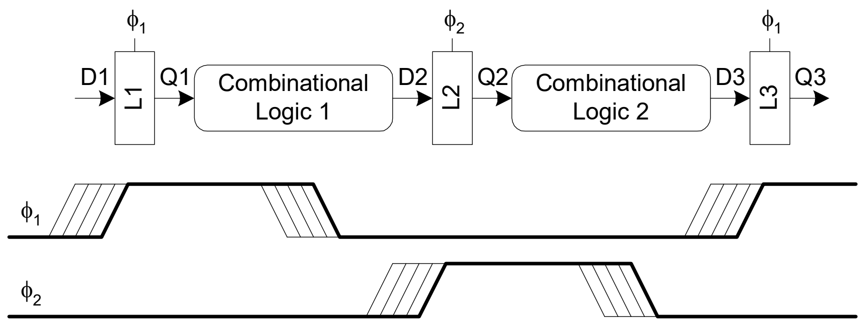
Skew: Latches

2-Phase Latches

$$t_{pd} \le T_c - \underbrace{\left(2t_{pdq}\right)}_{\text{sequencing overhead}}$$

$$t_{cd1}, t_{cd2} \ge t_{\text{hold}} - t_{ccq} - t_{\text{nonoverlap}} + t_{\text{skew}}$$

$$t_{\text{borrow}} \le \frac{T_c}{2} - \left(t_{\text{setup}} + t_{\text{nonoverlap}} + t_{\text{skew}}\right)$$



Pulsed Latches

$$t_{pd} \le T_c - \underbrace{\max\left(t_{pdq}, t_{pcq} + t_{\text{setup}} - t_{pw} + t_{\text{skew}}\right)}_{\text{sequencing overhead}}$$

$$t_{cd} \ge t_{\text{hold}} + t_{pw} - t_{ccq} + t_{\text{skew}}$$

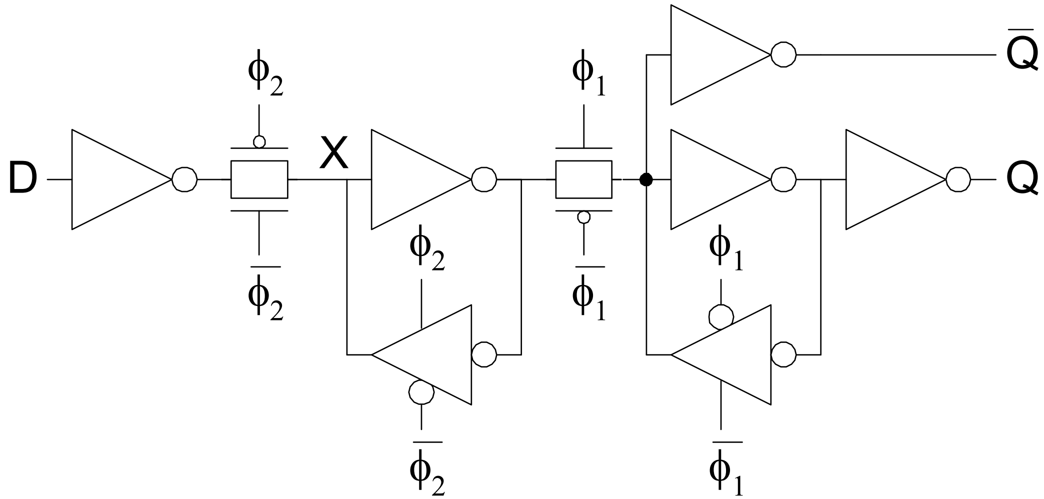
$$t_{\text{borrow}} \le t_{pw} - \left(t_{\text{setup}} + t_{\text{skew}}\right)$$

Two-Phase Clocking

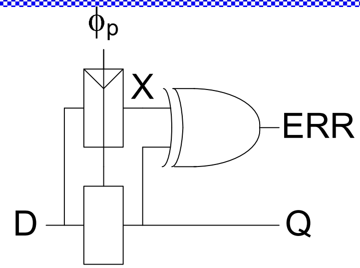
- If setup times are violated, reduce clock speed
- If hold times are violated, chip fails at any speed
- In this class, working chips are most important
  - No tools to analyze clock skew
- An easy way to guarantee hold times is to use 2-phase latches with big nonoverlap times
- Call these clocks ϕ_1, ϕ_2 (ph1, ph2)

Safe Flip-Flop

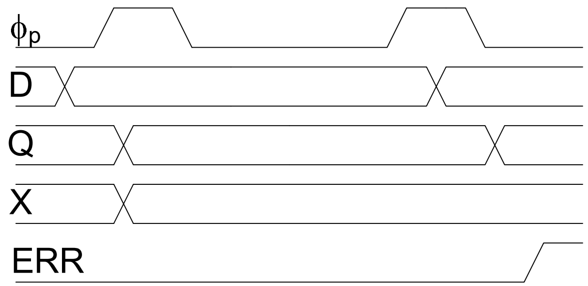
- Past years used flip-flop with nonoverlapping clocks
  - Slow – nonoverlap adds to setup time
  - But no hold times
- In industry, use a better timing analyzer
  - Add buffers to slow signals if hold time is at risk



Adaptive Sequencing



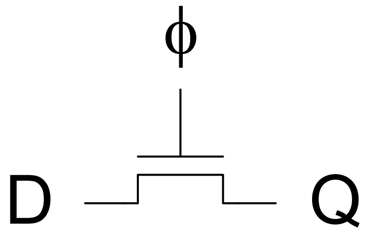
- Designers include timing margin
  - Voltage
  - Temperature
  - Process variation
  - Data dependency
  - Tool inaccuracies



- Alternative: run faster and check for near failures
  - Idea introduced as "Razor"
    - Increase frequency until at the verge of error
    - Can reduce cycle time by ~30%

Latch Design

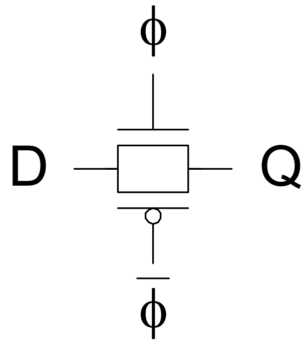
- Pass Transistor Latch
- Pros
  - +
  - +
- Cons
  - –
  - –
  - –
  - –
  - –
  - –



Used in 1970's

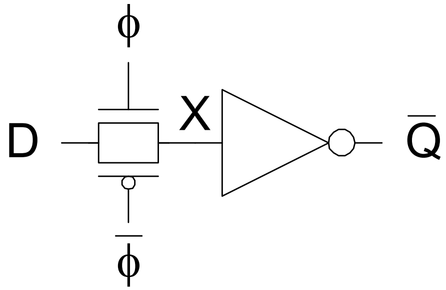
Latch Design

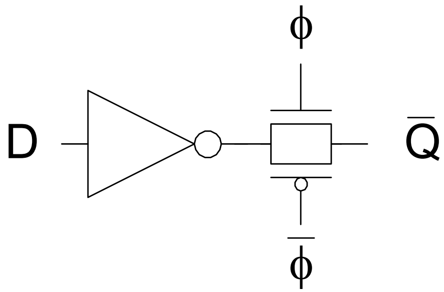
- Transmission gate
  - +
  - -



Latch Design

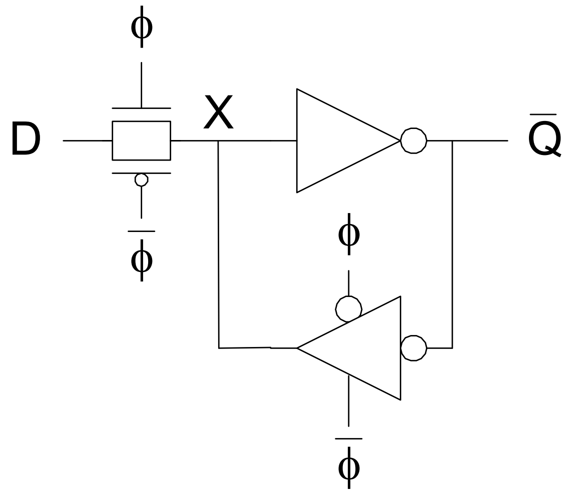
- Inverting buffer
  - +
  - +
  - + Fixes either
    - •
    - •
  - –





Latch Design

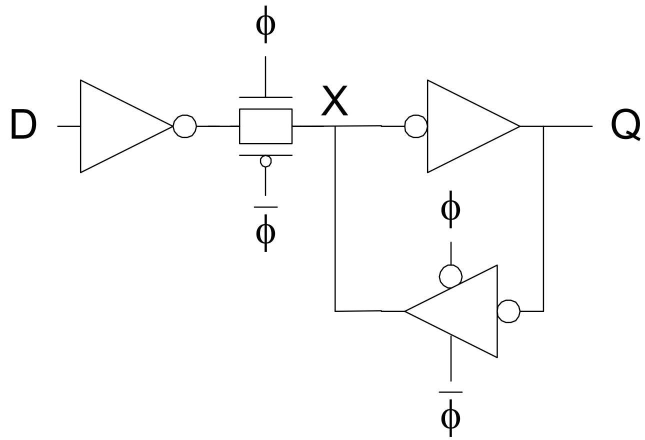
- Tristate feedback
  - +
  - –



- Static latches are now essential because of leakage

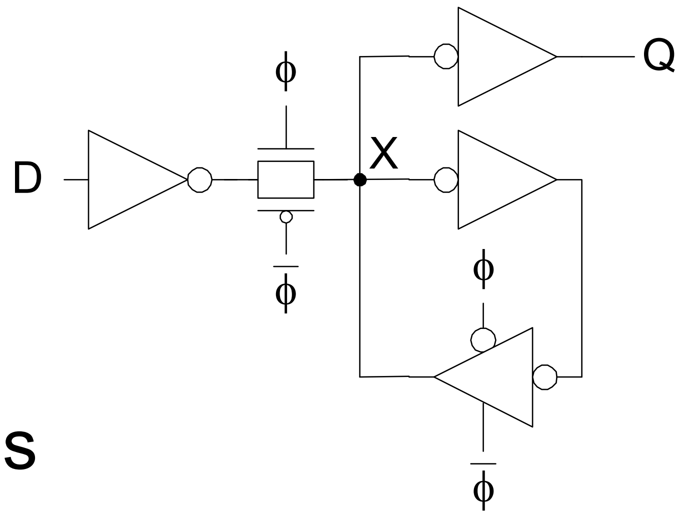
Latch Design

- Buffered input
  - +
  - +



Latch Design

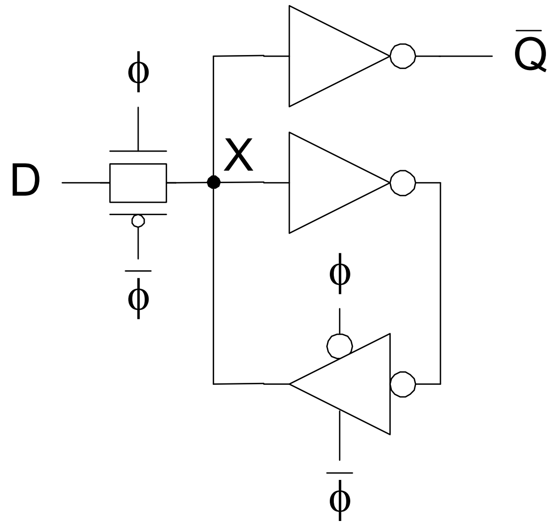
- ❑ Buffered output
  - +



- ❑ Widely used in standard cells
  - + Very robust (most important)
  - - Rather large
  - - Rather slow (1.5 – 2 FO4 delays)
  - - High clock loading

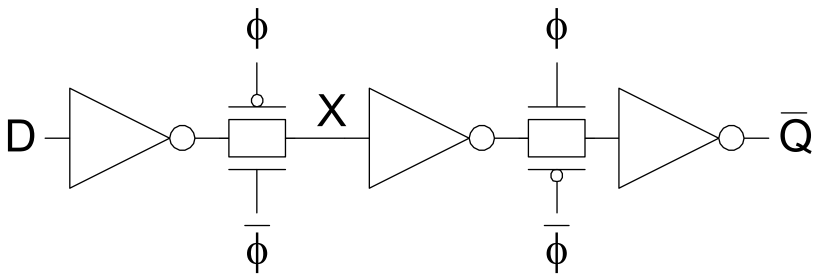
Latch Design

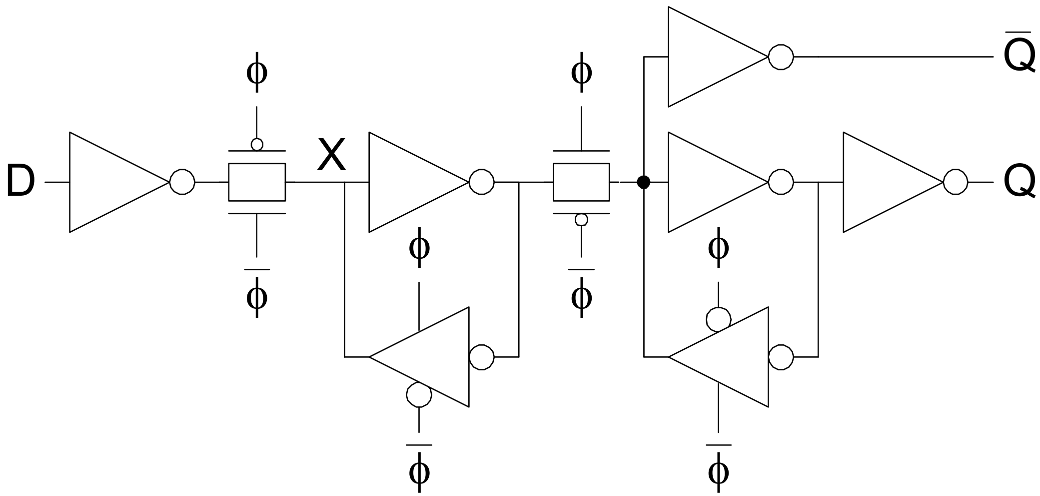
- Datapath latch
  - +
  - +
  - -



Flip-Flop Design

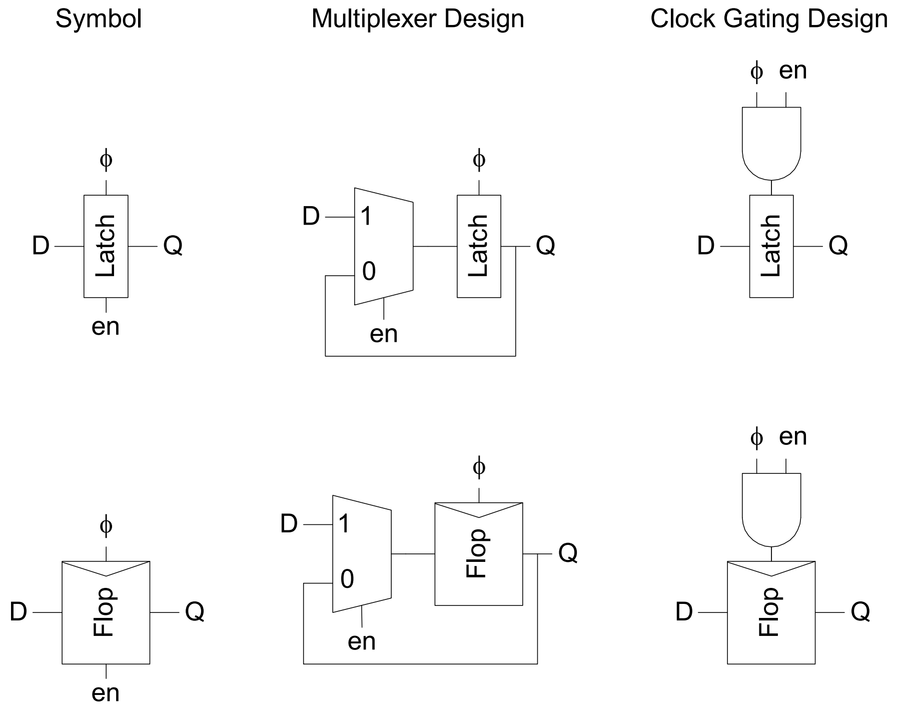
- Flip-flop is built as pair of back-to-back latches





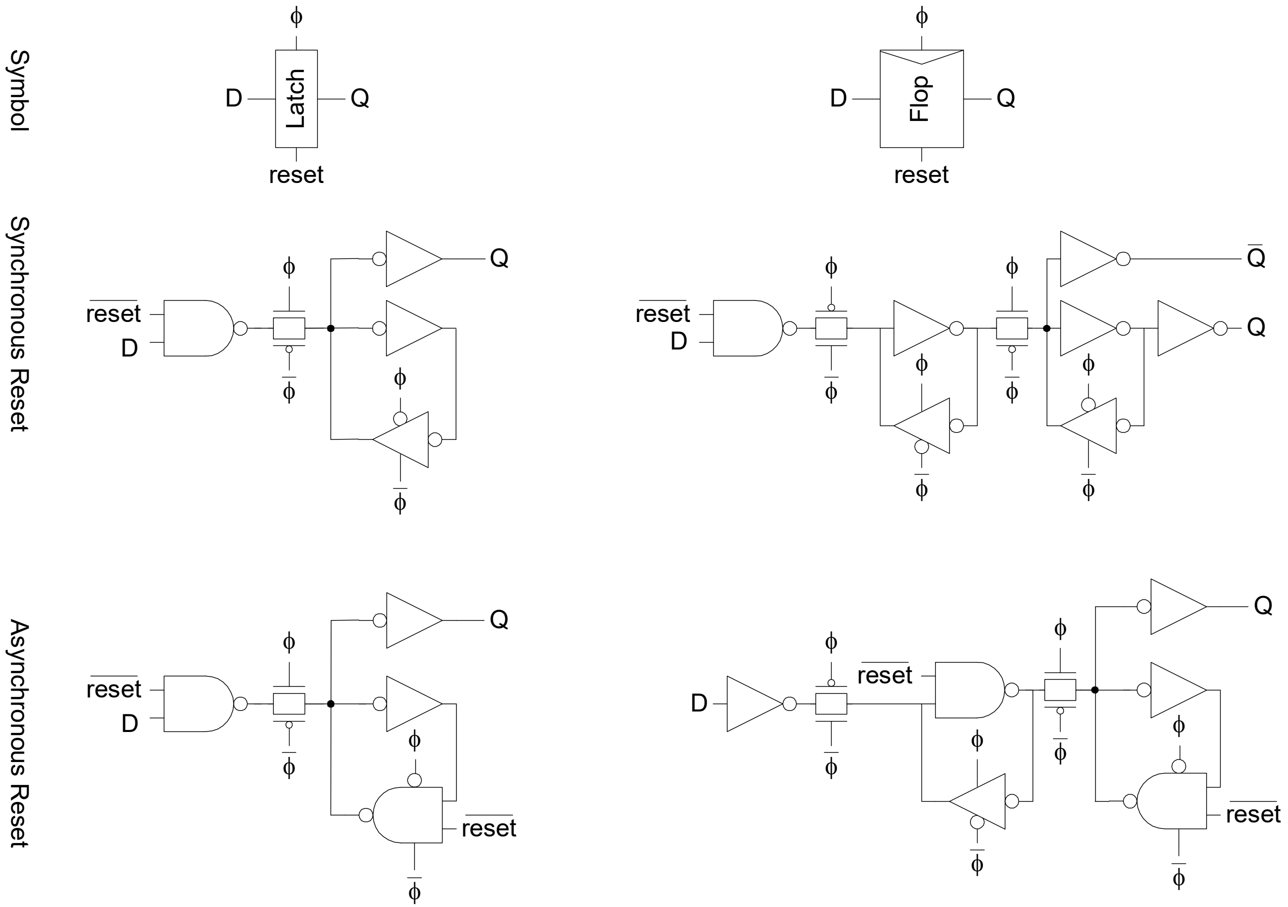
Enable

- Enable: ignore clock when en = 0
  - Mux: increase latch D-Q delay
  - Clock Gating: increase en setup time, skew



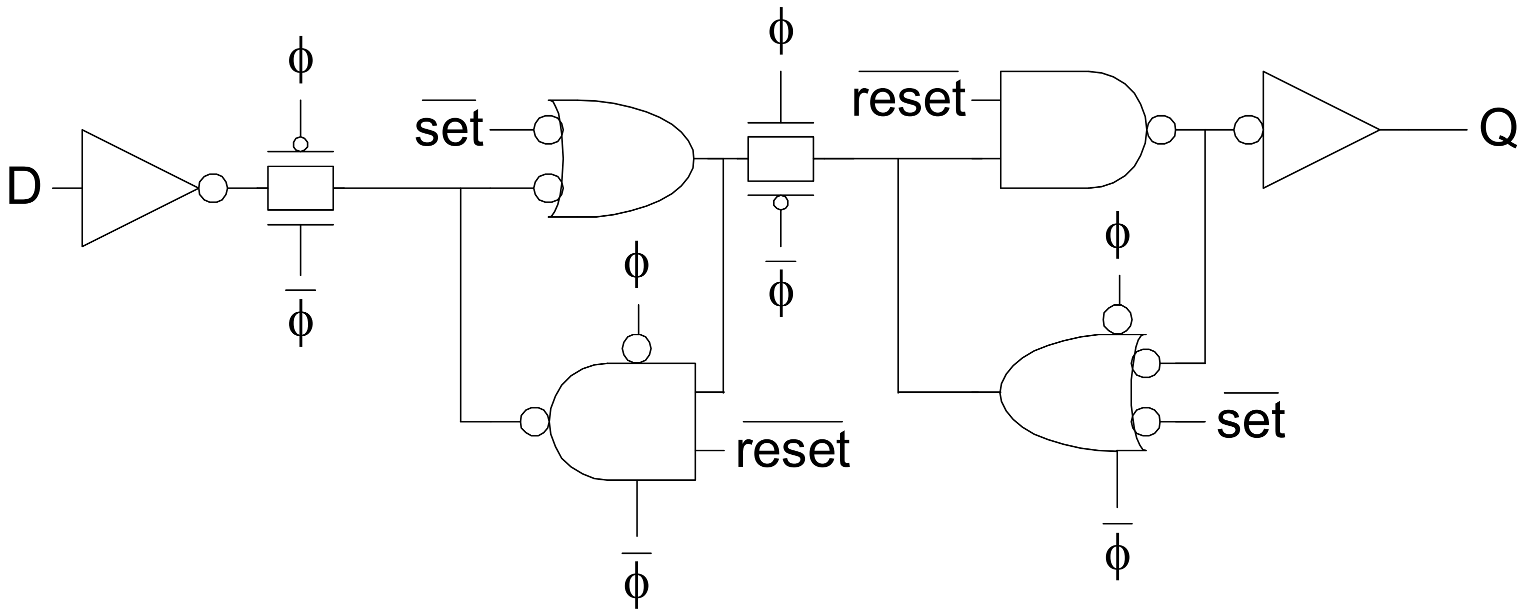
Reset

- Force output low when reset asserted
- Synchronous vs. asynchronous



Set / Reset

- Set forces output high when enabled
- Flip-flop with asynchronous set and reset



Summary

- Flip-Flops:
  - Very easy to use, supported by all tools
- 2-Phase Transparent Latches:
  - Lots of skew tolerance and time borrowing
- Pulsed Latches:
  - Fast, some skew tol & borrow, hold time risk

	Sequencing Overhead $(T_c - t_{pd})$	Minimum Logic Delay t_{cd}	Time Borrowing t_{borrow}
Flip-Flops	$t_{pcq} + t_{setup} + t_{skew}$	$t_{hold} - t_{ccq} + t_{skew}$	0
Two-Phase Transparent Latches	$2t_{pdq}$	$t_{hold} - t_{ccq} + t_{nonoverlap} + t_{skew}$ in each half-cycle	$T_c/2 - (t_{setup} + t_{nonoverlap} + t_{skew})$
Pulsed Latches	$\max(t_{pdq}, t_{pcq} + t_{setup} - t_{pw} + t_{skew})$	$t_{hold} - t_{ccq} + t_{pw} + t_{skew}$	$t_{pw} - (t_{setup} + t_{skew})$

Review

1. Distinguish combinational circuits and sequential circuits?
2. What are tokens in sequential circuits? Why are tokens needed?
3. What are propagation delay and contamination delay?
4. Explain setup time, hold time.
5. Explain max-delay constraints, min-delay constraints, time borrowing, and clock skew.
6. What are pros and cons of latches using pass transistors, transmission gates, inverting buffers, three state feedback, buffered input, datapath?